Measure Theoretic Conditional Expectation in an Elementary Setting

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Overview

The measure theoretic approach to conditional expectation can be confusing when compared to the traditional approach. In what follows we go through a conditional expectation problem within a discrete and familiar setting in an attempt to reduce this confusion. One of the most confusing/disturbing ideas that one must confront from the very definition of conditional expectation is the notion of non-measurable functions. We work through an explicit example where we demonstrate that a given function is not measurable in the "conditional" σ-algebra.

We can think of the conditional expectation of a function, f, on a set, Λ, as a weighted average of f over Λ.

From this perspective, you can think of the conditional expectation as giving the "best" representation of f with respect to a coarse σ-algebra, given that f lives on a finer one. Just as the "best" representation of an image at a larger pixel scale (coarse mesh) would be an average over the smaller scale pixels (refined mesh) that make up the larger pixel.

Elementary Probability Example

Let us start with a simple example. We have a die with six faces, $D_1, D_2, D_3, D_4, D_5, D_6$. Any roll of the die will produce one of these faces with an equal probability. Let's further suppose that the value of each side is the number of dots on the die. Now suppose that when the die is rolled, we don't see the die, but someone tells us partial information – only that the die is "even" or "odd".

Question: Given that we only know the even or oddness of the die, what is the "best" guess of the die value? Let's try to answer this question using the measure theoretic definition of conditional expectation.

Let $X = \{D_1, D_2, D_3, D_4, D_5, D_6\}$ and define a function P by $P(D_i) = \frac{1}{6}$, for $i \in \{1, 2, 3, 4, 5, 6\}$. The intent is that P will become a probability measure for the space we construct. Let $\mathcal{E} = 2^X$ be the σ-algebra consisting of the power set of X. We extend P for every element in the σ-algebra. Since the σ-algebra consists of all sets we need an assignment for an arbitrary set, A. The assignment is $P(A) = \frac{|A|}{6}$; that is, the cardinality of the set divided by 6. We now have a measure space; in fact, a probability space: $(X, \mathcal{E}, P)$. Note that for a probability space we need a sample space, X, a σ-algebra of sets (in the discrete case just an algebra) – whose elements are the *events* – and a function P which takes elements of the σ-algebra to [0,1] with the following properties:

- $P(X) = 1$.
- $P(\emptyset) = 0$.
- $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \quad \text{when } A_k \cap A_j = \emptyset \quad k \neq j$;

We now consider a random variable from which we will get a sub σ-algebra. Let

$$g(D_i) = \begin{cases} 0 & \text{if } i \text{ is even;} \\ 1 & \text{if } i \text{ is odd.} \end{cases}$$

In general, the σ-algebra generated from g is the algebra generated by preimages of Borel sets in $(-\infty, \infty)$. In the discrete case this reduces to the algebra of sets generated from $g^{-1}(a)$, $a \in (-\infty, \infty)$. Clearly, the interesting sets come from the values 0 and 1, all other values lead to the empty set. Consequently,

$$\mathcal{F} = \{\emptyset, \{D_1, D_3, D_5\}, \{D_2, D_4, D_6\}, \{D_1, D_2, D_3, D_4, D_5, D_6\}\}$$

In a discrete space a function, f, is measurable with respect to a σ-algebra if $f^{-1}(a)$ is an element in the σ-algebra for all $a \in (-\infty, \infty)$. This has implications for the σ-algebra $\mathcal{F}$.

Claim: Any function, f, which is measurable over $\mathcal{F}$ has the property that f is constant on the sets $\{D_1, D_3, D_5\}$ and $\{D_2, D_4, D_6\}$. More generally, we claim that any measurable function, f, in a given σ-algebra must be constant on the *minimal* elements of the algebra – elements which have no non-trivial subsets.†

To see this suppose that $f(D_1)$ differs from $f(D_3)$. Then the set, $A = \{D_1, D_3, D_5\} \cap f^{-1}(f(D_1))$, is a *strict*, non-trivial subset of $\{D_1, D_3, D_5\}$. This is true since A, by construction, is a subset of $\{D_1, D_3, D_5\}$; A contains D_1; and A does not contain D_3. Since $\mathcal{F}$ is a σ-algebra and A is the intersection of the $\mathcal{F}$-measurable sets $\{D_1, D_3, D_5\}$ and $f^{-1}(f(D_1))$, A must be in $\mathcal{F}$. However, we know that $\{D_1, D_3, D_5\}$ has no strict non-trivial subset in $\mathcal{F}$ – contradiction. Therefore, our premise that $f(D_1)$ and $f(D_3)$ could take differing values is incorrect. The same argument shows that $f(D_1)$ and $f(D_5)$ do not differ. One can repeat the above argument to show that f is constant on the other minimal set $\{D_2, D_4, D_6\}$.

Notice that while any function over the measure space $(X, \mathcal{E}, P)$ is measurable, we can write down a specific function that is non-measurable with respect to $\mathcal{F}$. We know that all we have to do is come up with a function that differs on either of the sets $\{D_1, D_3, D_5\}$ or $\{D_2, D_4, D_6\}$. For instance, the function: $f(D_i) = i$, for $i \in \{1, 2, 3, 4, 5, 6\}$, is a non-measurable function in $\mathcal{F}$.

Conditional Expectation

Given a probability space $(X, \mathcal{E}, P)$, the conditional expectation of an $\mathcal{E}$-measurable, P-integrable function f with respect to a sub σ-algebra $\mathcal{F}$ is the *unique $\mathcal{F}$-measurable* function labeled, $\mathbb{E}[f \mid \mathcal{F}]$, such that‡

$$\int_\Lambda \mathbb{E}[f \mid \mathcal{F}] \, dP = \int_\Lambda f \, dP \quad \forall \Lambda \in \mathcal{F} \tag{1a}$$

Let us write this again in, perhaps, an unusual way.

$$\int_\Lambda \mathbb{E}[f \mid \mathcal{F}] \, dP_{\mathcal{F}} = \int_\Lambda f \, dP \quad \forall \Lambda \in \mathcal{F} \tag{1b}$$

Equation (1b) is the same equation as (1a), but with the measure on the left written as $P_{\mathcal{F}}$ – the restriction of P to the sub σ-algebra $\mathcal{F}$. As a number, $P_{\mathcal{F}}(\Lambda) = P(\Lambda)$ for every $\Lambda \in \mathcal{F}$; the subscript is a reminder that the integrand on the left is required to be $\mathcal{F}$-measurable for the left-hand integral to be well-defined.

† Specifically, in a discrete setting, a set Z in a σ-algebra, $\mathcal{H}$, is *minimal* in $\mathcal{H}$ if there is no non-empty, strict subset, Y, of Z with $Y \in \mathcal{H}$.

‡ That such a unique function exists is a consequence of the Radon-Nikodym theorem.

Although it seems that f – itself – satisfies (1a), we have to be careful. The right-hand side of (1b) integrates an $\mathcal{E}$-measurable function against P, while the left-hand side requires an $\mathcal{F}$-measurable integrand against $P_{\mathcal{F}}$. Consequently, while f seems like a natural candidate for the conditional expectation, it is not necessarily $\mathcal{F}$-measurable. Typically, f is *non-measurable* with respect to the sub σ-algebra, $\mathcal{F}$. However, if f is $\mathcal{F}$-measurable then, by the definition above, it is its own conditional expectation.

The modern theory of probability – including conditional expectation – relies on measure theory which has its roots in Lebesgue measure theory – used to provide an alternative to the theory of Riemann Integration. From Lebesgue Measure Theory one is introduced to measurable and non-measurable sets/functions. The problem, in the context of Lebesgue, is that non-measurable sets/functions exist but specific examples are hard to come by. However, non-measurable functions appear *implicitly* in the definition of conditional expectation in that if they didn't the definition wouldn't be of any interest.

If one started out learning measure theory from the Lebesgue theory, the definition of conditional expectation might make readers somewhat uneasy as we are confronted with non-measurable sets from the very start. From this context, it is harder to get an intuitive idea of conditional expectation.

In the next section we compute the measure theoretic conditional expectation of a discrete function. In the process, we provide an intuitive idea of the measure-theoretic definition of conditional expectation which coincides with the traditional approach – except in the "singular" case which we discuss in the "Redux" section of this paper.

Example Calculation of Conditional Expectation

Consider the function from the second section on an elementary die example, $f(D_i) = i$, which is measurable in the space $(X, \mathcal{E}, P)$. Let $\mathcal{F}$ be the sub σ-algebra of that section. We now compute the conditional expectation of f with respect to $\mathcal{F}$. Using (1b) we choose the *minimal* sets: $\Lambda_1 = \{D_1, D_3, D_5\}$ and $\Lambda_2 = \{D_2, D_4, D_6\}$. Since the conditional expectation function is constant on each of these sets, equation (1b) will provide a way to find the value on each set. Since the union of Λ_1 and Λ_2 constitutes the entire set, X, we will know the value of the conditional expectation function on all of X; hence we will know the conditional expectation function.

Proceeding, we have

$$\int_{\Lambda_1} \mathbb{E}[f \mid \mathcal{F}]\, dP_{\mathcal{F}} = \int_{\Lambda_1} f\, dP \tag{2}$$

and

$$\int_{\Lambda_2} \mathbb{E}[f \mid \mathcal{F}]\, dP_{\mathcal{F}} = \int_{\Lambda_2} f\, dP \tag{3}$$

We know $\mathbb{E}[f \mid \mathcal{F}]$ is *constant* on Λ_1. We label this value as $\mathbb{E}[f \mid \mathcal{F}](\Lambda_1)$. From (2) we have

$$\begin{aligned}
\int_{\Lambda_1} \mathbb{E}[f \mid \mathcal{F}]\, dP_{\mathcal{F}} &= \int_{\Lambda_1} f\, dP \\
\mathbb{E}[f \mid \mathcal{F}](\Lambda_1) \int_{\Lambda_1} dP_{\mathcal{F}} &= \int_{\Lambda_1} f\, dP \\
\mathbb{E}[f \mid \mathcal{F}](\Lambda_1) \cdot P_{\mathcal{F}}(\Lambda_1) &= \int_{\Lambda_1} f\, dP = f(D_1) \cdot P(D_1) + f(D_3) \cdot P(D_3) + f(D_5) \cdot P(D_5) \qquad (**)
\end{aligned}$$

$$
\begin{aligned}
\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_1) &= \frac{\int_{\Lambda_1} f\,dP}{P_{\mathcal{F}}(\Lambda_1)} = f(D_1)\cdot\frac{P(D_1)}{P(\Lambda_1)} + f(D_3)\cdot\frac{P(D_3)}{P(\Lambda_1)} + f(D_5)\cdot\frac{P(D_5)}{P(\Lambda_1)} \\
\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_1) &= \frac{\int_{\Lambda_1} f\,dP}{P_{\mathcal{F}}(\Lambda_1)} = 1\cdot\frac{\frac{1}{6}}{\frac{1}{2}} + 3\cdot\frac{\frac{1}{6}}{\frac{1}{2}} + 5\cdot\frac{\frac{1}{6}}{\frac{1}{2}} \\
\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_1) &= \frac{\int_{\Lambda_1} f\,dP}{P_{\mathcal{F}}(\Lambda_1)} = 1\cdot\frac{1}{3} + 3\cdot\frac{1}{3} + 5\cdot\frac{1}{3} \\
\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_1) &= 3
\end{aligned}
$$

Likewise, $\mathbb{E}\left[f \mid \mathcal{F}\right]$ is constant on the set Λ_2. As we did above, we find the value of $\mathbb{E}\left[f \mid \mathcal{F}\right]$ on the set Λ_2. As with Λ_1, label $\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_2)$ as the constant value of $\mathbb{E}\left[f \mid \mathcal{F}\right]$ on Λ_2. We have

$$
\begin{aligned}
\int_{\Lambda_2} \mathbb{E}\left[f \mid \mathcal{F}\right]\,dP_{\mathcal{F}} &= \int_{\Lambda_2} f\,dP \\
\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_2)\int_{\Lambda_2} dP_{\mathcal{F}} &= \int_{\Lambda_2} f\,dP \\
\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_2)\cdot P_{\mathcal{F}}(\Lambda_2) &= \int_{\Lambda_2} f\,dP = f(D_2)\cdot P(D_2) + f(D_4)\cdot P(D_4) + f(D_6)\cdot P(D_6) \\
\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_2) &= \frac{\int_{\Lambda_2} f\,dP}{P_{\mathcal{F}}(\Lambda_2)} = f(D_2)\cdot\frac{P(D_2)}{P(\Lambda_2)} + f(D_4)\cdot\frac{P(D_4)}{P(\Lambda_2)} + f(D_6)\cdot\frac{P(D_6)}{P(\Lambda_2)} \\
\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_2) &= \frac{\int_{\Lambda_2} f\,dP}{P_{\mathcal{F}}(\Lambda_2)} = 2\cdot\frac{\frac{1}{6}}{\frac{1}{2}} + 4\cdot\frac{\frac{1}{6}}{\frac{1}{2}} + 6\cdot\frac{\frac{1}{6}}{\frac{1}{2}} \\
\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_2) &= \frac{\int_{\Lambda_2} f\,dP}{P_{\mathcal{F}}(\Lambda_2)} = 2\cdot\frac{1}{3} + 4\cdot\frac{1}{3} + 6\cdot\frac{1}{3} \\
\mathbb{E}\left[f \mid \mathcal{F}\right](\Lambda_2) &= 4
\end{aligned}
$$

Since a function is determined once we know what its values are on every $x \in X$, we have found the conditional expectation function, $\mathbb{E}\left[f \mid \mathcal{F}\right]$, as we know its values on every $x \in X$.†

From equation $(**)$ we see that the value of the function $\mathbb{E}\left[f \mid \mathcal{F}\right]$ evaluated on any 'x' value in a minimal set, Λ, is‡

$$
\mathbb{E}\left[f \mid \mathcal{F}\right](x) = \frac{\int_{\Lambda} f\,dP}{P(\Lambda)} = \int_{\Lambda} f\,dP_{\Lambda} \tag{4}
$$

That is, the value of the conditional expectation on any minimal set is the *weighted average* of f over the minimal set. The weights are determined by normalizing the probability measure P over the minimal set.

This is the view that was promised at the beginning of this paper.

Example Redux

Let's adjust the die example probabilities and redo the calculation of the conditional expectation. Replace the uniform P from before with the new probability measure:

$$
P(A) = \frac{|A \cap \{D_1, D_3, D_5\}^c|}{3} \quad A \in \mathcal{E}
$$

† In the "Redux" section we examine more closely what "knowing the value on every x" means.

‡ The notation, P_{Λ} means that we have normalized the measure, P, so that $P_{\Lambda}(\Lambda) = 1$. To do this we are *assuming* that $P(\Lambda) \neq 0$.

That is, the probability of a set, A, is the number of dice in A that are not in the set $\{D_1, D_3, D_5\}$ divided by 3. With this definition, $P(\Lambda_1) = 0$ and $P(\Lambda_2) = 1$.

Given P, we can find the value of $\mathbb{E}[f \mid \mathcal{F}]$ on the set Λ_2 as before using equation (4). However, we can't use it to find the value $\mathbb{E}[f \mid \mathcal{F}]$ on the set Λ_1 as $P(\Lambda_1) = 0$.

It turns out, we don't have to determine the values with any specificity on Λ_1. We can *set* the values of $\mathbb{E}[f \mid \mathcal{F}]$ on Λ_1 to 0; or, to any other *single* value for that matter†. The reason for this is that the definition of conditional expectation determines a unique element in the function space on $\mathcal{F}$. However, each element in such a space is actually an equivalence class of measurable functions over $\mathcal{F}$ that differ on a set of measure 0.

We can't directly compute what the value of a representative of $\mathbb{E}[f \mid \mathcal{F}]$ is on the set Λ_1, but we don't have to, since it's a set of measure 0, any value on this set will give us a function that is in the equivalence class of the conditional expectation element. Therefore, we may take $\mathbb{E}[f \mid \mathcal{F}]$ to be 0 on Λ_1. This gives us a representative for the conditional expectation as we have values for all $x \in X$. Through this representative we can produce the associated equivalence class of functions. Consequently, we know $\mathbb{E}[f \mid \mathcal{F}]$ in the function space $L^1(X, \mathcal{F}, P)$.

In the case of singular sets (a set of probability 0) we may revise equation (4) and write that on minimal sets, $\mathbb{E}[f \mid \mathcal{F}]$ is constant and satisfies:

$$\mathbb{E}[f \mid \mathcal{F}](x) \cdot P(\Lambda) = \int_{\Lambda} f \, dP \tag{5}$$

Note that when $P(\Lambda) = 0$ both sides of (5) vanish identically, so (5) imposes no constraint on the value of $\mathbb{E}[f \mid \mathcal{F}]$ on Λ – consistent with the freedom (afforded by the equivalence-class viewpoint) to assign that value arbitrarily.

Here, with a measure theoretic methodology, we have an elegant way to talk about conditional expectation in a singular context that doesn't exist in the traditional approach.

† We know that Λ_1 is minimal; so, for the conditional expectation function to be measurable it must have the same value on all elements of Λ_1.

APPENDIX

A Discrete Conditional Expectation Result

We generalize the discussion from the previous sections and work with functions on discrete sets with a given algebra of sets.

Definition: Given a discrete set X and an algebra of sets $\mathcal{F}$, a set A is minimal in X with respect to $\mathcal{F}$ iff $A \in \mathcal{F}$ and no non-empty proper subset of A is contained in $\mathcal{F}$.

Let X be a set with a finite number of elements. For any given σ-algebra, $\mathcal{F}$, (in this case algebra) of sets of X, let $M_{\mathcal{F}} = \{M_i\}_{i=1}^N$ be the set of all *distinct* minimal sets of X in $\mathcal{F}$. The set $M_{\mathcal{F}}$ exists as we are dealing with a finite set X with a finite power set 2^X. The set $M_{\mathcal{F}}$ is formed simply by examining each element in the power set of X and adding it to $M_{\mathcal{F}}$ if it is minimal in $\mathcal{F}$.

Claim: The set $M_{\mathcal{F}}$ is a *partition* of the set X. Meaning:

- $\forall i, j \in \{1, \ldots, N\} \quad M_i \cap M_j = \emptyset \quad (i \neq j)$;
- $X = \bigcup\limits_{i=1}^{N} M_i$.
- The algebra generated by the sets in $M_{\mathcal{F}}$ is the collection of the unions of each element in the power set of $M_{\mathcal{F}}$. Further, the algebra generated is $\mathcal{F}$.

We prove the first property by contradiction. Suppose $M_i \neq M_j$ are both minimal in $\mathcal{F}$ with $M_i \cap M_j \neq \emptyset$. Since $\mathcal{F}$ is an algebra, $M_i \cap M_j \in \mathcal{F}$. It is a non-empty subset of M_i, so by minimality of M_i it must equal M_i; hence $M_i \subseteq M_j$. By minimality of M_j, this in turn gives $M_i = M_j$, contradicting $M_i \neq M_j$.

We prove the second property by contradiction. If the union of minimal sets did not span X, then the set, $Z_0 = \left\{ \bigcup\limits_{i=1}^{N} M_i \right\}^c$, is non-empty. The set Z_0 cannot be minimal as we have already accounted for all minimal sets in our collection. Set $\mathcal{H} = 2^{Z_0} \setminus \{\emptyset, Z_0\}$ – the power set of Z_0 less Z_0 and $\emptyset$. Note that for Z_0 to be non-minimal, it cannot consist of 1 element. Therefore, $\mathcal{H}$ is a non-empty collection of sets. If none of the non-empty sets are in $\mathcal{F}$, then Z_0 is minimal – a contradiction. Therefore, there must be a non-trivial strict subset of Z_0, Z_1, that is in $\mathcal{F}$. We can repeat this argument with Z_1 to produce, Z_2, $\ldots Z_i$ producing Z_{i+1}, etc; noting, that each time this is done the number of elements in the current "Z set" decreases by at least 1. Consequently, the process cannot proceed indefinitely. Let m be the step where this process fails; that is, there is no strict subset, Z_m, of set Z_{m-1} which is a subset of $\mathcal{F}$. But this means that Z_m is minimal by definition – contradiction. Consequently, the second property holds.

The third property follows as none of the $\{M_i\}_{i=1}^N$ can produce sets of finer granularity than themselves; they behave as if they were single elements. Therefore, the only way to produce new sets is by union, and consequently, the collection of all elements generated is the collection of the union of each element from the power set of $\{M_i\}_{i=1}^N$. Finally, if we label $\mathcal{G}$ as the algebra generated by $M_{\mathcal{F}}$, then since its generator sets, $\{M_i\}_{i=1}^N$ are elements of $\mathcal{F}$, it must be the case that $\mathcal{G} \subseteq \mathcal{F}$. To show equality, suppose that there is a set $F \in \mathcal{F}$ but $F \notin \mathcal{G}$. But $F = F \cap X = F \cap \bigcup\limits_{i=1}^{N} M_i = \bigcup\limits_{i=1}^{N} (F \cap M_i)$. Since for a given i, M_i is minimal in $\mathcal{F}$, and $F \in \mathcal{F}$, the intersection must be either M_i or $\emptyset$. Consequently, F is a union of an element of the power set of $\{M_i\}_{i=1}^N$, which has been shown to be a member of $\mathcal{G}$. Consequently, $\mathcal{G} = \mathcal{F}$.

Let $(X, \mathcal{E}, P)$ be a probability space with X discrete, let f be measurable on it, and let $\mathcal{F}$ be a sub σ-algebra of $\mathcal{E}$. We examine the expectation of the conditional expectation, $\mathbb{E}\left[f/\mathcal{F}\right]$.†

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† Remember, the conditional expectation is not a number; it is a random variable (a measurable function on

$$
\begin{aligned}
E[\mathbb{E}\,[f/\mathcal{F}]] &= \int_X \mathbb{E}\,[f/\mathcal{F}]\; dP_{\mathcal{F}} = \int_X f\,dP & (6)\\
&= \int_{\overset{N}{\underset{i=1}{\cup}} M_i} f\,dP \\
&= \sum_{i=1}^{N} \int_{M_i} f\,dP \\
&= \sum_{i=1}^{N} \int_{M_i} \mathbb{E}\,[f/\mathcal{F}]\; dP_{\mathcal{F}} \\
&= \sum_{i=1}^{N} \mathbb{E}\,[f/\mathcal{F}](M_i)\, P(M_i) & (7)
\end{aligned}
$$

In equation (6) we have the following identity: $E[\mathbb{E}\,[f/\mathcal{F}]] = E[f]$. From equation (7) we have that the expectation of the conditional expectation of a σ-algebra generated by minimal sets is just the weighted sum of the values of the conditional expectation on these sets.

Finally, if g is a discrete function, the collection of all non-empty sets of the form $g^{-1}(a)$, $a \in (-\infty, \infty)$, is a finite collection of sets, $\{M_i\}_{i=1}^{N}$, for some positive integer N.‡ The collection of the union of each element from the power set of these sets generates an algebra, $\mathcal{F}$, with each M_i ($1 \le i \le N$) being the minimal sets in this algebra. We write $\mathbb{E}\,[f \mid g]$ to mean $\mathbb{E}\,[f/\mathcal{F}]$. Consequently, we may write the above as:

$$
\begin{aligned}
E[\mathbb{E}\,[f \mid g]] &= \int_X \mathbb{E}\,[f \mid g]\; dP = \int_X f\,dP & (6')\\
&= \int_{\overset{N}{\underset{i=1}{\cup}} M_i} f\,dP \\
&= \sum_{i=1}^{N} \int_{M_i} f\,dP \\
&= \sum_{i=1}^{N} \int_{M_i} \mathbb{E}\,[f \mid g]\; dP \\
&= \sum_{i=1}^{N} \mathbb{E}\,[f \mid g]\,(M_i)\, P(M_i) & (7')
\end{aligned}
$$

Here, the term, $\mathbb{E}\,[f \mid g]\,(M_i)$ is interpreted as the value of $\mathbb{E}\,[f/\mathcal{F}]$ on *any* element of the set M_i – as the value of the function $\mathbb{E}\,[f/\mathcal{F}]$ is constant on M_i. We also drop the $dP_{\mathcal{F}}$ notation, which was only used as a crutch in the previous part of the paper. Lastly, we know from equation (5) how to compute the values of $\mathbb{E}\,[f \mid g]$ for all M_i – even when the probability of a "g" event is 0.

Equation (6') says that $E[\mathbb{E}\,[f \mid g]] = E[f]$; that is, the expectation of the conditional expectation of f is the same as the expectation of f. Equation (7') states that on each level set of g, $M_i = g^{-1}(a_i)$, the conditional expectation $\mathbb{E}\,[f \mid g]$ is the P-weighted average of f over M_i; summing those values weighted by $P(M_i)$ recovers $E[f]$.

$(X, \mathcal{F})$). As such, it too has an expectation.

‡ The N sets correspond to N distinct values $\{a_i\}_{i=1}^{N}$, with each $M_i = g^{-1}(a_i)$ and each a_i a real number.